Seeing what you eat

When should a nutrition app refuse to answer?

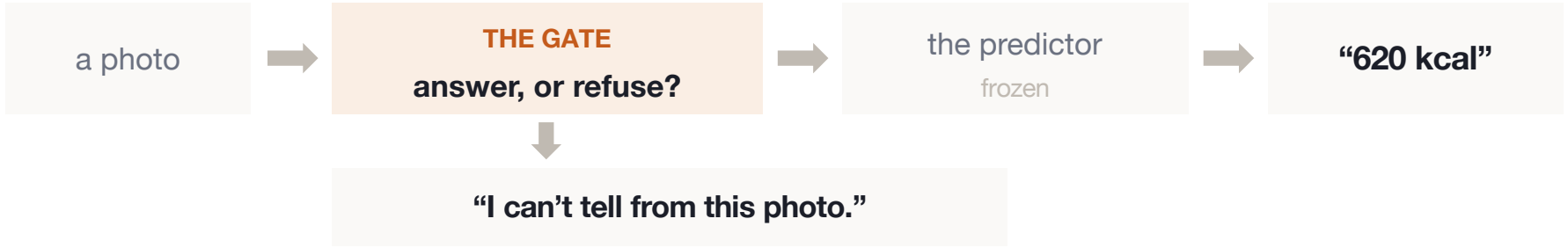
COMPSCI 760 · Group 1 · Research Project Update: Methodology and Results

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Code and results: **github.com/sushantkrishnan/seeing-what-you-eat**

RECAP

Every app answers every photo. Should it always?



THE PROBLEM

Calorie apps give one number for every photo, and they never decline.

WHAT WE PROPOSED

Keep the predictor fixed, and benchmark the signals an app could use to decide whether to answer at all, measured between refusing at random and a perfect refuser.

OBJECTIVES · STATUS

O1 Benchmark
Which signals predict when the answer is wrong?
done

O2 Ceiling
How much is on the table, and on which axis?
done

O3 Ranges
Does the best signal give ranges worth having?
first cut done

Which signals let an app know when it is guessing, and can any of them run on the phone?

WHY THE GATE, NOT THE PREDICTOR

The headroom is in refusing, not in predicting.

OUR OWN NUMBERS, 507 TEST DISHES, CALORIE ERROR IN KCAL [OURS]

71

when we answer every photo

our frozen predictor

50

if a perfect refuser declined one photo in ten

That is 20 kcal of headroom; our best real gate takes 11 of them, landing at 60 (slide 10).

87

if we named and weighed the ingredients instead

with the 22% mass error we measure, and one ingredient in five misnamed, which we assume [ours, oracle budget]

The best published number, 41 kcal, needed a depth channel [1], which most phone photos do not have. The published colour-only result is 70.6 kcal [1].

A perfect refuser is a ceiling, and a real gate already takes half of the room under it. That is why we froze the predictor and study the gate.

SINCE THE PROPOSAL

What the markers asked, and what we changed.

FEEDBACK WE RECEIVED, AND WHAT WE DID WITH IT

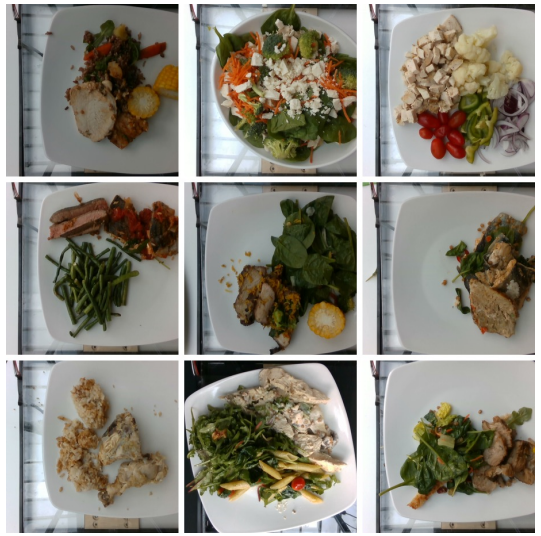
Freezing the predictor and studying the gate is the right design.	We kept it. Nothing on the predictor side has changed.
Which evaluation metrics, exactly?	One fixed set, shown on slide 7: error at a fixed answer rate, area under the curve with the share of the gap it closes, rank correlation, and what each gate refuses. We then score every gate a second time with meal size held fixed (slide 11).
How will you show a difference is significant?	Every number carries a 95% interval from a bootstrap over plate sessions, and every gate is tested against refusing at random, paired, with a Holm correction (slides 10 and 11).

CHANGES WE MADE, AND WHY

Calibration is now carved by plate session, not by dish.	98% of calibration dishes shared a session with a fit dish. We fixed that, and the predictor moved from 73.4 to 70.9 kcal, inside its interval → slide 5
We report 90% answered and show the whole curve.	That is where an app would sit. The proposal quoted 50%, where every gate looks better → slide 7
Every gate is scored again inside size bands.	A control that reads only the predicted size tied our best gate. Holding size fixed separates reading the dish from reading its size → slide 11
Confidence ranges started three weeks early.	The first cut is done, and it found the next problem: large meals → slide 13

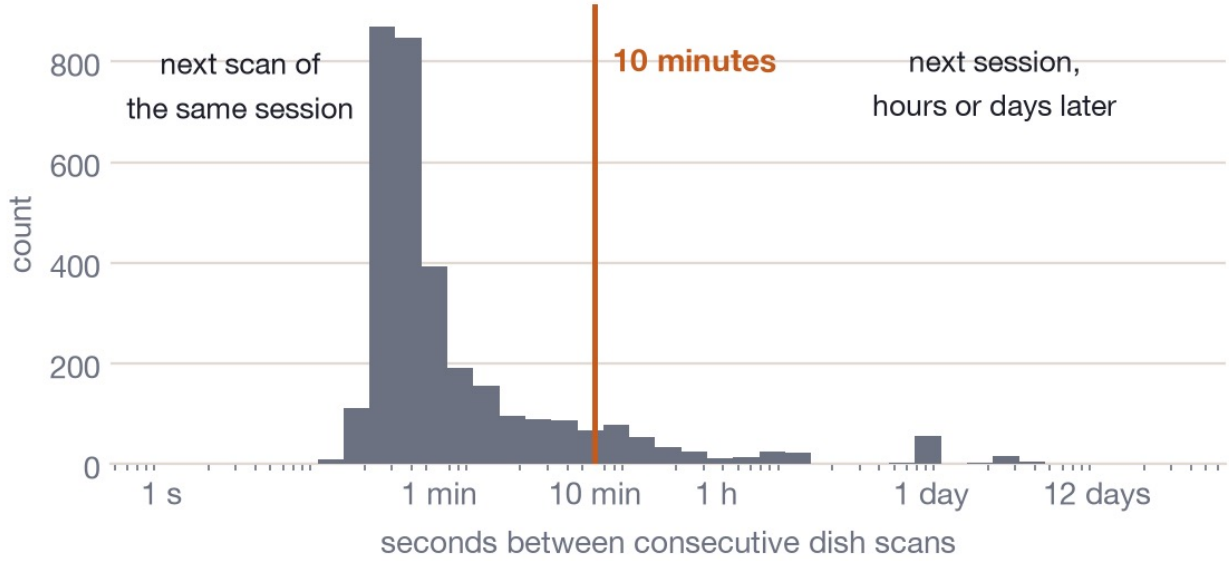
DATASET

507 test dishes are 127 plate sessions.



Nutrition5k [1]: overhead photos from one rig, with every ingredient weighed. That weighed truth is what lets us build a perfect refuser to compare against.

SCANS OF ONE PLATE ARRIVE 40 SECONDS APART



Errors within a session are correlated (intraclass correlation 0.34), so the session is our unit for the calibration carve and for every interval in this talk.

SPLITS

4,768	3,490	2,755 / 507	1,924 / 831	367 / 127
official split ids [1]	have overhead RGB	train / test, official	fit / calibration, by session	sessions, train / test

EXPERIMENTAL DESIGN

One frozen model, one linear head, one tuned knob.



THE ONE TUNED KNOB

The ridge penalty, alpha, chosen on the calibration split from a 25-point quarter-decade grid. It picked 5,623.

FIXED IN ADVANCE

64 PCA components for the distance gate, 32 ensemble heads, and 10 nearest neighbours.

ONE RESULTS FILE

8 damage levels × 507 dishes × 3 backbones = 12,168 rows. Every result in this talk is a lookup on that file.

COMPUTE

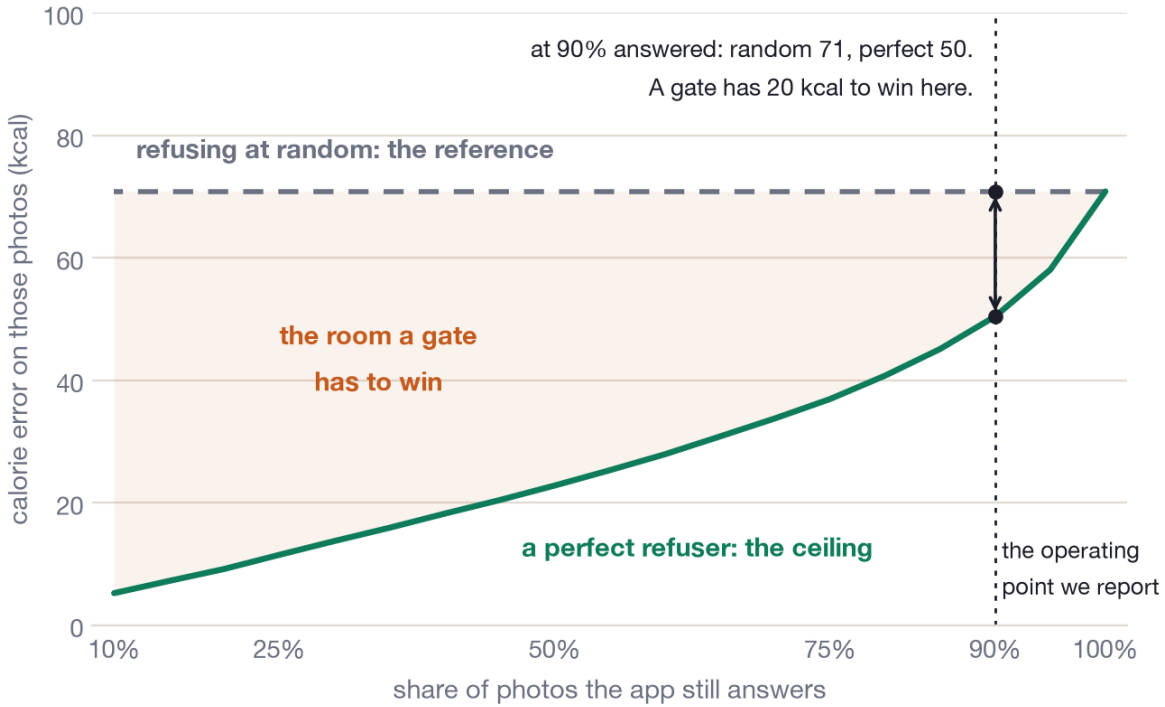
One M-series laptop at about 68 images per second through the backbone. The whole pipeline reruns in under 15 minutes.

There is no training loop, so there is no convergence curve to show, and nothing was tuned on the test split.

Leak discipline: the heads are fit on the fit dishes; their errors are measured on the calibration dishes, which they never saw and which share no session with them; the error head is trained on those errors; and everything is reported on the 507 test dishes.

HOW WE SCORE A GATE

How to read every chart that follows.



Clean test photos, CLIP [ours]; protocol from [3] and [6]. Random is a reference, not a bound.

A GATE IS SCORED ON FOUR THINGS

1 Error at 90% answered

The MAE on the photos it still answers, at the operating point. We show the whole curve as well.

2 Area under the curve

AURC [11], and the share of the random-to-perfect gap the gate closes across all answer rates.

3 Rank correlation

Spearman ρ between the signal and the true error.

4 What it refuses

The mean size of the meals it keeps, and the share of 400+ kcal meals it refuses, compared with the perfect refuser. Ours.

EVERY NUMBER CARRIES AN INTERVAL we resample the 127 plate sessions 2,000 times and recompute every curve on the same resample, so the tests are paired. Each gate is tested against refusing at random and Holm-corrected across the ten candidates. Every gate is then scored twice more: with meal size held fixed (slide 11) and on relative error (slide 12).

EXISTING RESULTS VS OURS: THE PREDICTOR

The published baseline, matched without training.



On the 507 official test dishes CLIP gives 70.9 kcal [62, 80], so the published 70.6 sits inside our interval. Portion mass: 43.8 g (22%).

Calories, MAE	kcal	source
RGB, Inception V2, trained	**70.6**	[1]
RGB-D	**47.6**	[1]
Volume scalar, best	**41.3**	[1]
Ours, CLIP, frozen	**70.9**	[ours, replicated]
Ours, SigLIP 2, frozen	**67.5**	[ours]
Ours, DINOv3, frozen	**63.2**	[ours]
2026 re-split, 60/15/25	**28.0**	not comparable

The 28.0 kcal paper re-splits Nutrition5k at random, so near-duplicate scans land on both sides. We keep [1]'s split and do not compare to it.

We matched the baseline, so the predictor stays frozen. Everything from here is about the gate.

RESULT 1, COMPLETED

Pixel signals see the damage, not the harm.

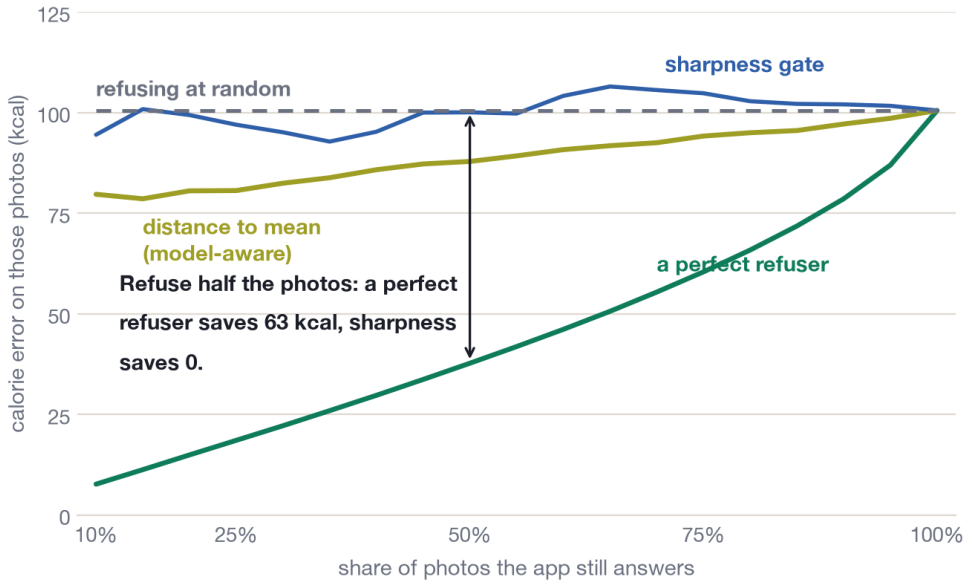
ONE DISH, FOUR PHOTOGRAPHS, ITS OWN NUMBERS [OURS]



clean	blurred	cropped	phone
sharpness **463**	sharpness **1.8**	sharpness **30**	sharpness **26**
mass error **1.0×**	mass error **1.5×**	mass error **7.9×**	mass error **3.0×**

Sharpness rates this cropped photo 16× better than the blurred one, yet its mass error is 8× the clean photo's and 5× the blurred one's, because the plate rim is the only clue to portion size. Across all 507 dishes the cropped rung is 4× sharper than the blurred rung and carries 2.3× its mass error. The phone rung is an imitation and costs 1.2× on calories.

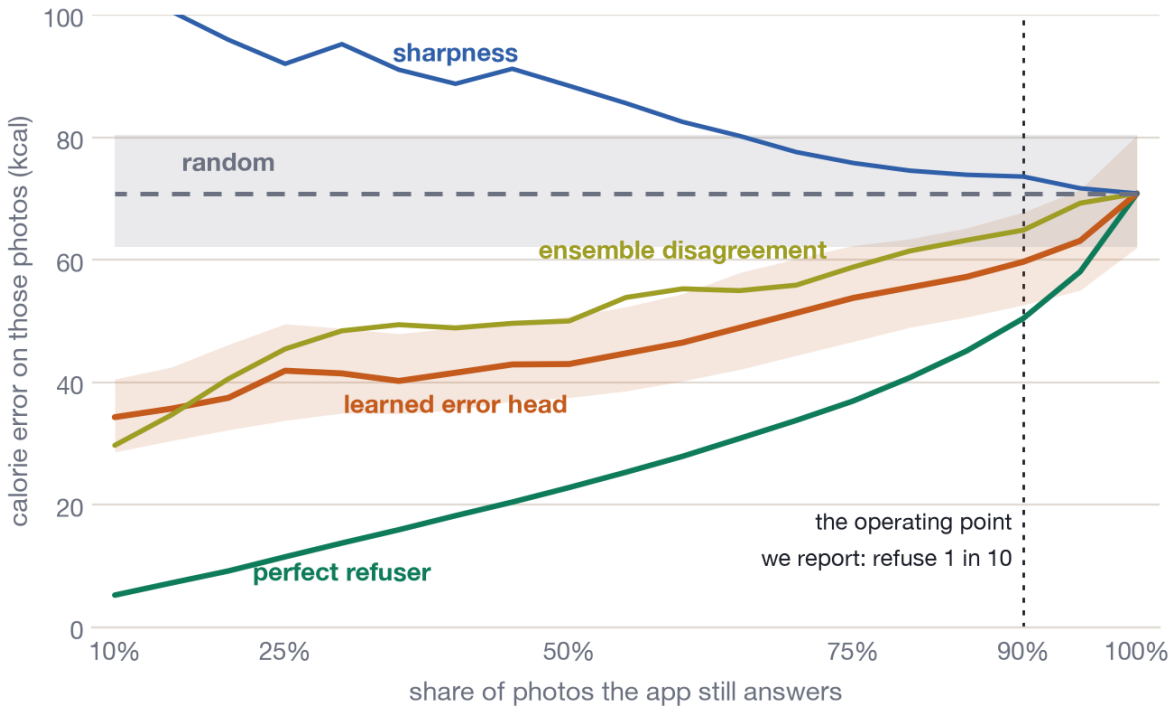
REFUSE THE WORST PHOTOS, 8 DAMAGE LEVELS [4] POOLED [OURS]



DETECTED, AND IT BARELY HELPS pixel metrics separate damaged photos from clean ones almost perfectly (AUC ≈ 1.00), the test [2] asks for. Yet over 4,056 dish × damage rows every pixel gate sits within 2 kcal of random at 90% answered, and sharpness is worse than random on all three backbones. Where a pixel gate does beat random it is dropping whole damage levels ($|\rho| \le 0.08$): never more than a quarter of the gap at one answer rate, and 9% of the area over the whole curve. On clean photos none beats random. The dish explains 62% of the error variance, the damage level 9%, their interaction 29%. A gate has to read the dish.

RESULT 2, COMPLETED

Model-aware signals bend the curve. So does size.



Clean test photos, CLIP [ours]. The bands are each curve's own 95% interval; the tests are paired, so the differences are tighter than the bands suggest (slide 11).

AT 90% ANSWERED, CLIP [OURS]

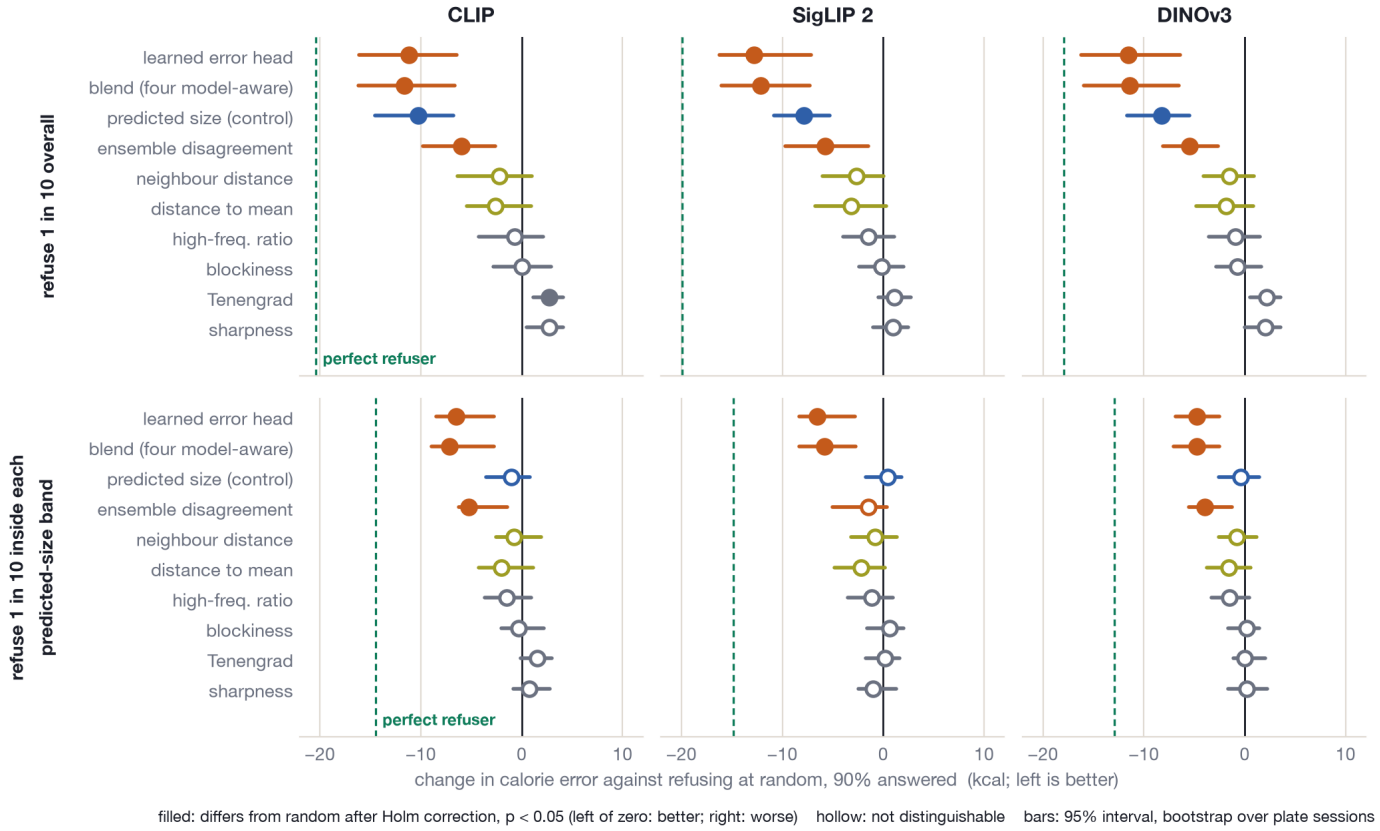
gate	kcal	vs random
perfect refuser	**50.4**	—
learned error head [10]	**59.7**	**–11.2 [–16, –6] ✓**
predicted size (control)	**60.6**	**–10.2 [–15, –7] ✓**
ensemble disagreement [9]	**64.9**	**–6.0 [–10, –3] ✓**
distance to mean [7]	**68.2**	–2.6 [–5, 1]
neighbour distance [8]	**68.6**	–2.2 [–6, 1]
random (answer all)	**70.9**	—
sharpness	**73.6**	+2.8 [0, 4]

✓ better and ✗ worse than random after Holm correction, $p < 0.05$. The control scores risk by the predicted calorie count alone. A blend of the four model-aware signals (slide 11) is not listed: it is the head (paired $p = 0.69$).

WHAT SURVIVED the learned error head beats random by 11 kcal at 90% answered ($p < 0.01$). So does the control, by 10, and the two are statistically tied (paired $p = 0.76$). Ensemble disagreement clears the correction at 6 kcal. Neither distance gate can be told apart from random here: they helped on the damage ladder (slide 9) because they detect damage, not difficulty.

RESULT 3, COMPLETED

Hold meal size fixed and half the win survives.



Inside a size band every gate keeps meals about the size random keeps (245–258 kcal against 256), so what survives is the dish, not its size.

THE CONTROL GOES TO ZERO

The predicted-size control goes from –10 / –8 / –8 kcal overall to –1.0 / +0.4 / –0.4 inside bands. That is by construction, and it shows the bands are doing their job.

THE HEAD KEEPS HALF

The learned error head goes from –11 / –13 / –12 overall to –6.5 / –6.5 / –4.7 inside bands, still clear of random on all three (Holm $p < 0.01$) and now ahead of the control (paired $p < 0.03$).

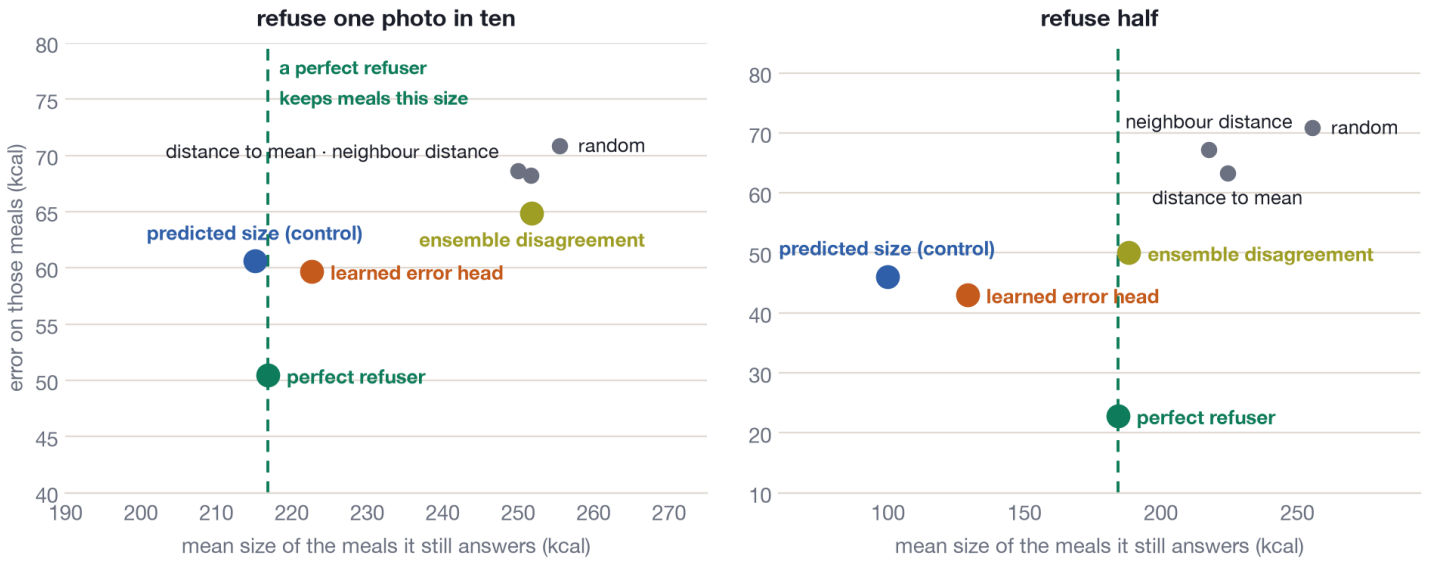
THE CEILING MOVES TOO

Even a perfect refuser goes from –20 / –20 / –18 to –14 / –15 / –13. A quarter to a third of the headroom at 90% was meal size.

CLIP / SigLIP 2 / DINOv3, 90% answered, change against refusing at random. Ten bands, with edges from the calibration split's predictions.

RESULT 3, COMPLETED

Absolute error rewards refusing big meals.



IS THE SIGNAL JUST THE SIZE? [OURS]

Spearman ρ with	size	error, in band
learned error head	**0.77**	**0.18**
predicted size (control)	**1.00**	**0.14**
ensemble disagreement	**0.41**	**0.19**
the true error itself	**0.41**	—

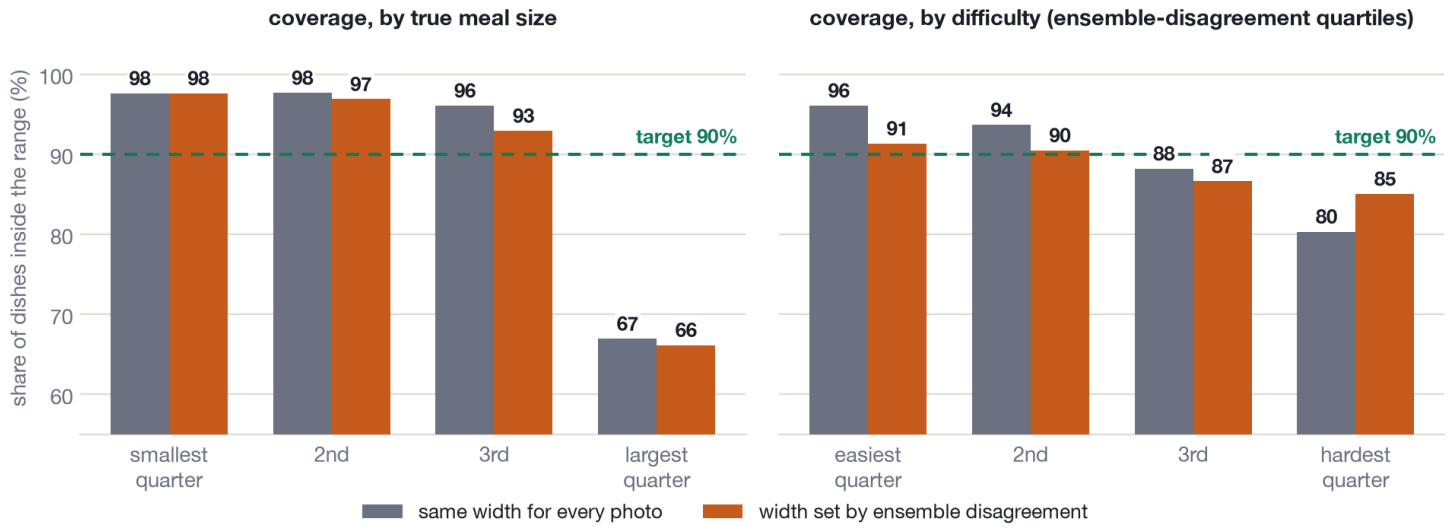
Spearman ρ on clean test photos, CLIP. The right column is the mean ρ with the true error inside predicted-size quartiles, a coarser cut than slide 11's ten bands, so some size remains in it.

TWO LOSSES, TWO ORACLES at 90% answered the absolute-error oracle refuses 34% of 400+ kcal meals and only 3% of small ones, and the head and the control follow it (they keep 223 and 215 kcal meals, against 256 overall). The relative-error oracle (|error| / meal size, with a 50 kcal floor) refuses 0% and 27%. Scored that way, no gate beats random at 90%: the control is worse (+1.6 points, $p < 0.01$), the head does not clear the correction (+1.2, $p = 0.05$), and only ensemble disagreement leans the right way (–2.7, $p = 0.11$).

Neither loss can judge a gate on its own. The within-band score on slide 11 can, and we report the kept-set size beside every error.

RESULT 4, FIRST CUT

Ranges: honest on average, not on large meals.



	plain	adaptive
coverage	**89.5%**	88.4%
interval	[86, 93]	[84, 92]
median half-width	±151 kcal	**±133 kcal**
largest meals	**67%**	66%
hardest quarter	80%	**85%**
within ±100 kcal	13%	**24%**
interval score	478	**446**

Split conformal [5], 90% target, 831 calibration dishes, 507 clean test photos, CLIP [ours]. The adaptive width is a normalised score [12] scaled by ensemble disagreement, which also cuts the difficulty quartiles.

Bold marks the better of the two on each row. The interval score is the width plus a penalty for misses; lower is better.

TWO THINGS THIS TELLS US (1) The 90% average hides 98% on the smaller meals and 67% on the largest quarter, and the adaptive width barely grows with size. It is the same size effect as on slides 11 and 12, and it is the next O3 problem.

(2) Cut by difficulty, the adaptive width moves coverage from the easiest quarter (96% → 91%) to the hardest (80% → 85%) and improves the proper score. As a check: calibrated on clean photos and deployed on damaged ones, coverage falls to 81%; recalibrated, it returns to 90% at 28% more width. That is a check, not a finding.

CHALLENGES, TIMELINE, ROLES

Progress against the proposal timeline.

TWO CHALLENGES, AND WHAT WE DID

Correlated test dishes

Errors cluster by plate session (ICC 0.34), so the session is now the unit of resampling and the calibration carve respects it.

A gate that cheats

Refusing big meals looks like skill. A control gate exposed that, so every gate is now scored again inside predicted-size bands, and we report the kept-set size beside every error.

Also: a rank-deficient covariance (2,304 dimensions on 1,924 dishes, so Mahalanobis runs on 64 components, fixed in advance), and no phone photos in the data (our phone rung is an imitation, and labelled as one).

PROPOSAL TIMELINE VS ACTUAL

planned	milestone	status
17 Aug	Pipeline, 3 backbones, first results	**done**
11 Sep	All gates scored, one results file	**done 17 Sep**
20 Sep	Literature review	**submitted 20 Sep**
11 Oct	Confidence ranges	**first cut 17 Sep**
25 Oct	Final report	**next**

Next: size-aware ranges for large meals (conformalised quantile regression); the within-band threshold as a deployable gate; the deployment mixture on all backbones; the costlier tier (test-time augmentation, a second backbone); and the report.

WHO OWNS WHAT

Nisarg
pipeline, results file, framing

Shreya
damage levels, pixel gates

Keanu
model-aware gates, curves

Sushant
literature, writing, data

Kevin
confidence ranges

CONCLUSIONS

What we know now that we did not in August.

1 **Photo quality is not the axis.**

We detect damage almost perfectly, yet it explains only 9% of the error while the dish explains 62%. On clean photos no pixel gate beats random at any answer rate, and on the damage ladder the most any captures over the whole curve is 9% of the area, by dropping whole damage levels.

2 **Half of the best gate's win is meal size. The other half is real.**

At 90% answered the learned error head cuts the error by 11 kcal against random, but so does simply reading the predicted size (10), and the two tie. Inside predicted-size bands the control drops to zero while the head keeps 5–7 kcal ($p < 0.01$ on three backbones).

3 **The loss decides which gate looks honest.**

Absolute error rewards refusing big meals; relative error rewards refusing small ones, and under it no gate beats random. Ensemble disagreement is the signal least tied to size, and it keeps most of its gain inside bands on two of three backbones.

4 **Ranges are honest on average, not on large meals.**

The largest quarter of meals is covered only 67% of the time, the same size effect again. Adaptive width moves coverage from the easiest quarter to the hardest. Size-aware ranges are O3's job by 11 October.

So far: an app can tell when it is guessing about a meal of a given size, worth about 6 kcal at one refusal in ten, and it already knows the size. The signals that need no model all fail; the ones that work are a linear head on features the app already computes.

Every method here is borrowed and cited. The benchmark, the size control, the sessions and the findings are ours.

CLOSING

Sources, AI use, and where the code is.

[1] Q. Thames, A. Karpur, W. Norris, F. Xia, L. Panait, T. Weyand, and J. Sim, “Nutrition5k: Towards automatic nutritional understanding of generic food,” in Proc. IEEE/CVF CVPR, 2021, pp. 8899–8907.

[2] B. Coburn, J. He, M. E. Rollo, S. S. Dhaliwal, D. A. Kerr, and F. Zhu, “Comprehensive evaluation of large multimodal models for nutrition analysis: A new benchmark enriched with contextual metadata,” arXiv:2507.07048, 2025 (extended version of the IEEE BHI 2025 paper).

[3] P. Grother and E. Tabassi, “Performance of biometric quality measures,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 29, no. 4, pp. 531–543, Apr. 2007.

[4] D. Hendrycks and T. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations,” in Proc. ICLR, 2019.

[5] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.

[6] Y. Geifman and R. El-Yaniv, “SelectiveNet: A deep neural network with an integrated reject option,” in Proc. ICML, PMLR 97, 2019.

[7] K. Lee, K. Lee, H. Lee, and J. Shin, “A simple unified framework for detecting out-of-distribution samples and adversarial attacks,” in Adv. Neural Inf. Process. Syst. 31, 2018.

[8] Y. Sun, Y. Ming, X. Zhu, and Y. Li, “Out-of-distribution detection with deep nearest neighbors,” in Proc. ICML, PMLR 162, 2022.

[9] B. Lakshminarayanan, A. Pritzel, and C. Blundell, “Simple and scalable predictive uncertainty estimation using deep ensembles,” in Adv. Neural Inf. Process. Syst. 30, 2017.

[10] C. Corbière, N. Thome, A. Bar-Hen, M. Cord, and P. Pérez, “Addressing failure prediction by learning model confidence,” in Adv. Neural Inf. Process. Syst. 32, 2019.

[11] Y. Geifman, G. Uziel, and R. El-Yaniv, “Bias-reduced uncertainty estimation for deep neural classifiers,” in Proc. ICLR, 2019.

[12] H. Papadopoulos, V. Vovk, and A. Gammerman, “Regression conformal prediction with nearest neighbours,” J. Artif. Intell. Res., vol. 40, pp. 815–840, 2011.

Re-split comparison on slide 8: V. Awasthi et al., Int. J. Intell. Eng. Syst., vol. 19, no. 2, 2026, doi:10.22266/ijies2026.0228.08. The interval score and the Holm correction are standard; full references in the report. All figures and tables are ours unless tagged [1]; images on slides 5 and 9 are Nutrition5k dishes [1], reproduced under the dataset licence.

GENERATIVE AI USE Claude Code (Anthropic) assisted with coding and planning: it helped write the pipeline and analysis code, the figures and the slide builders, and helped draft the talk script, under the group’s direction. The research question, the experimental design, the checking of every number against the results files, and the conclusions are the group’s own.

CODE AND RESULTS **github.com/sushantkrishnan/seeing-what-you-eat** · every result in this deck is regenerated from results/ by the deck builder; the oracle budget and the variance split reproduce from oracle_ladder.py and error_sources.py